

Detection and Characterisation of Subsurface Ice in Lunar South Polar Regions

Using Chandrayaan-2 DFSAR Compact Polarimetry

Instrument: Chandrayaan-2 DFSAR (Dual-Frequency SAR)

Frequency: L-band 1.25 GHz (S-band 2.50 GHz)

Polarimetry: Compact polarimetry — full Stokes reconstruction

Ice criterion: $CPR > 0.8$ AND $DOP < 0.13$ (dual threshold)

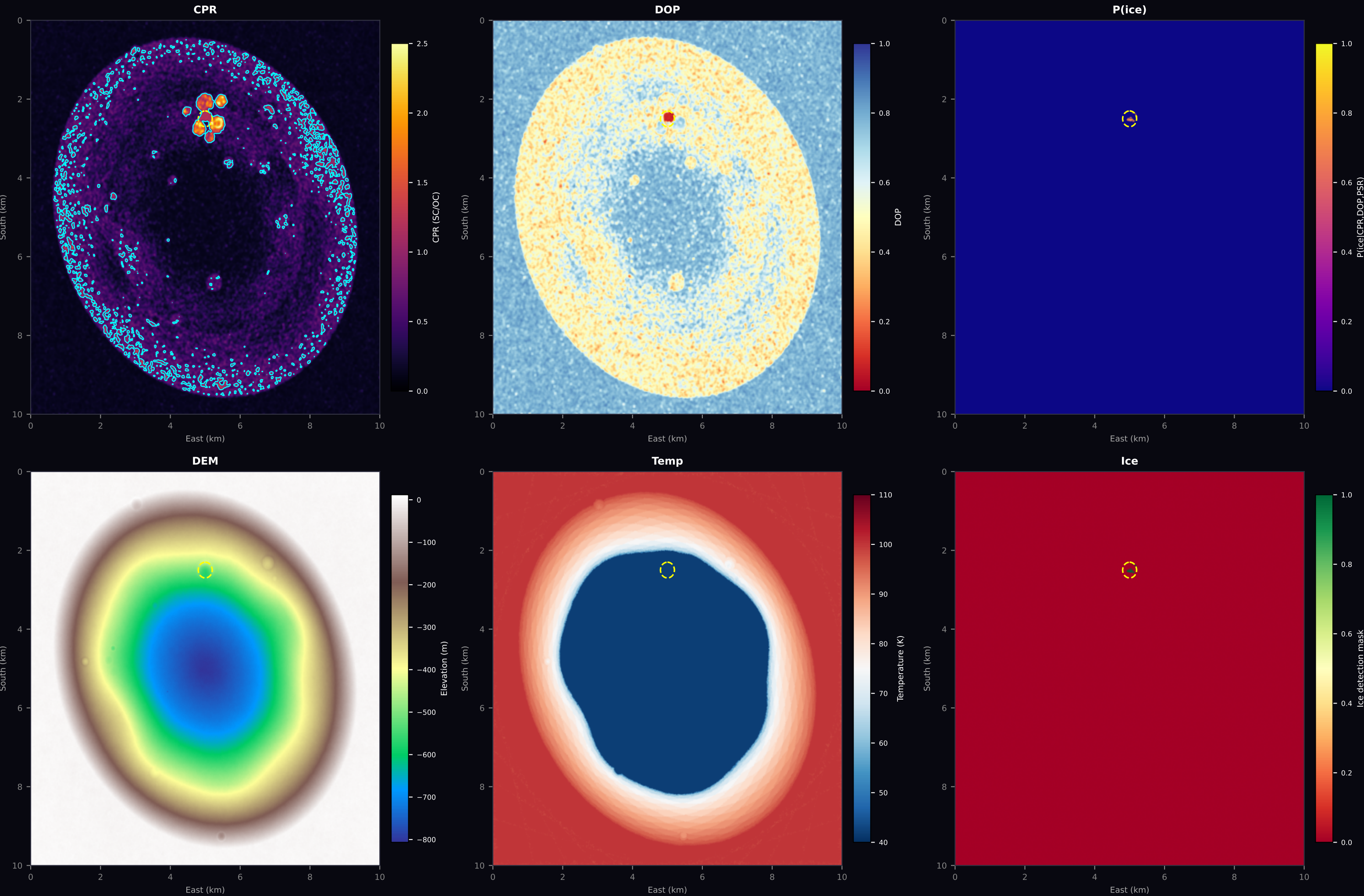
Target: Faustini Doubly Shadowed Crater (PSR), 87°S

L-band pen. depth: ~4.5 m in dry regolith

Analysis: m-chi decomposition · Bayesian posterior · Monte Carlo

Scene resolution: 10 m/pixel | 10 km × 10 km coverage

Scene Overview — Chandrayaan-2 DFSAR | Faustini PSR



Ice Detection & Uncertainty Quantification

